RODRIGO MENDES LIMA

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INTRODUCTION

First-year Computer Engineering Master's student at the University of Minho, pursuing specializations in Applications Engineering and Intelligent Systems. My primary focus lies in leveraging intelligent systems to process and manipulate data, while applying solid software engineering practices to develop and structure the surrounding core system architecture. I am highly motivated to bridge the gap between data-driven AI processing and robust, traditional application development.

EDUCATION

University Of Minho *2022 - 2025*
BSc in Computer Engineering Final Grade: 16/20

University Of Minho *2025 - Present*
MSc in Computer Engineering Final Grade: -/20
Specializations: *Applications Engineering and Intelligent Systems*

SOFT SKILLS

Key Competencies

- Team collaboration and communication
- Analytical thinking and problem solving
- Responsibility and continuous learning

Languages

- Portuguese (Native)
- English (Fluent)

TECHNICAL SKILLS

Programming Languages	C, C++, Python, Java, JavaScript
Back-End	Spring Boot, Node.js
Databases	PostgreSQL, MySQL, MongoDB
Front-End	Vue.js, HTML5, TailwindCSS
Data & AI	Scikit-learn, PyTorch, Pandas, RAG, LLMs, Embedding
DevOps	Docker, Kubernetes, Google Cloud, Git, GitHub, Ansible

RESEARCH & PROFESSIONAL INTERESTS

While my recent experience has spanned full-stack web development and applied deep learning research, my primary drive lies in **intelligent systems integration, data-driven application development, and robust software architecture**. Moving forward, I am deeply interested in exploring how to leverage artificial intelligence to process and manipulate complex data, while applying solid engineering practices to build the surrounding core systems. My goal is to design and develop smart, scalable software solutions that seamlessly bridge the gap between advanced AI capabilities and traditional, reliable application architectures.

PROJECTS

Ticketing System *Project in Applications Engineering*

- Developing an application to manage the ticketing process for a public transport network.
- Directly involved in the system's architectural design using **UML models** and logical architecture diagrams (Spring Boot-style & Patterns).
- Directly involved in the configuration of **Spring Boot**, managing the modularity of its components and overall system architecture.
- Led the development of the user interface, ensuring a seamless experience, and conducted comprehensive functional testing.
- Engineered robust backend and frontend integration, prioritizing maintainability and clean code practices.

Intelligent System for Air Quality Prediction *Project in Intelligent Systems*

- Led the numerical data extraction ans treatment pipeline from **Open-Meteo** and **Copernicus Sentinel-5P** sources, and initiated the development of the **AirSight** web interface.
- The project utilizes a multi-source data pipeline, integrating atmospheric and climatic variables from **Copernicus Sentinel-5P**, **Sentinel-2**, and the **Open-Meteo API** to predict air quality.
- Employs a Multi-Agent System (MAS) architecture built with **LangGraph** to orchestrate independent agents tasked with optimizing hyperparameters for deep learning models.
- Integrates state-of-the-art computer vision architectures, including **ResNet, VGG, ViT, and Swin-T**, with a RAG-based explanation module using **ChromaDB** and **LLMs** for context-aware analysis.

PostgreSQL Database Optimization *Database Administration*

- Analyzed transactional queries to systematically improve database performance.
- Optimized **SQL code**, implemented strategic **Indexes**, and fine-tuned **system parameters** and parallelism settings to enhance execution efficiency.
- Conducted extensive **benchmarking** within Google Cloud virtual machine environments to ensure the reproducibility and reliability of performance results.

Cloud Deployment of AirTrail Webapp *Cloud Computing Applications and Services*

- Implemented automated cloud deployment pipelines using **Ansible** and **Kubernetes** (GKE).
- Monitored system state using **Google Cloud Monitoring** tools and conducted performance benchmarking with **Apache JMeter**.
- Optimized system performance and availability under high traffic by configuring **auto-scaling** and **load balancing** solutions.

Personal Optimization Sensing System *Sensing and the Environment*

- Developed an intelligent health optimization platform that aggregates physiological and performance data from **Strava, Fitbit, and Continuous Glucose Monitors (CGM)**.
- The Project integrates predictive models using **Random Forest** to analyze injury risk based on fatigue, daily activity levels, and sleep patterns.
- Includes a **Retrieval-Augmented Generation (RAG)** pipeline utilizing **Ollama (LLM)** and **ChromaDB** to generate dynamic, personalized training and nutritional plans based on validated scientific literature.
- Architected a modular infrastructure with a **Vue.js** frontend, centralizing data management on **Firebase Firestore** for scalable multi-user support.